

Water

ONE OXYGEN ATOM AND TWO HYDROGEN ATOMS BOND TO FORM THE COMPOUND WATER.

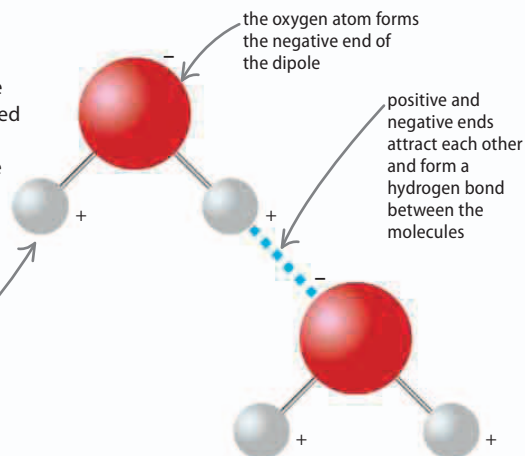
Water is one of the very few natural substances that are liquid in everyday conditions. Its unusual properties stem from the oxygen atom in its molecules.

Hydrogen bonds

In addition to the covalent bonds that join the hydrogen and oxygen atoms in a water molecule, there are bonds between the molecules themselves. The electrons in the covalent bond are pulled closer to the oxygen atom than to the hydrogen atoms. This makes the oxygen atom negatively charged and the hydrogen atoms positive. These opposite charges on different molecules attract each other in what is known as a hydrogen bond.

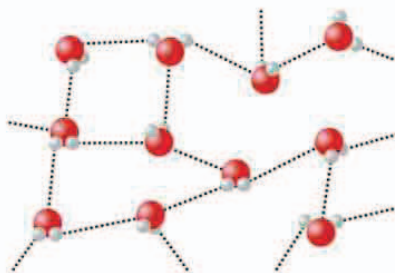
▷ **Dipole** The areas of charge in the water molecule are called poles. A hydrogen bond is formed between the negative end of one molecule and the positive end of another.

each hydrogen atom forms the positive end of the dipole



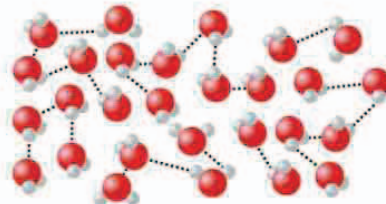
States of water

On Earth, water is mainly liquid—it covers 70 percent of the planet's surface. However, the other states of water are just as familiar—polar regions are covered in ice, while the atmosphere is filled with water vapor. Like all gases, water vapor has a lower density than water. However, almost uniquely among natural substances, when water freezes into ice (solid) it expands and has a lower density. As a result, ice floats on water. With other substances, their solid states have higher densities and sink under their liquid states.



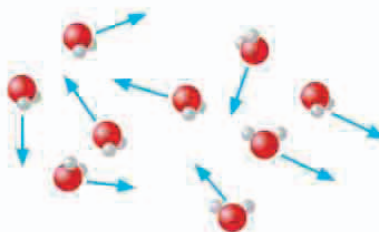
◁ Ice

An ice crystal is held together with hydrogen bonds. As the crystal forms, the molecules spread out so each molecule can bond with three others. This makes the molecules take up a larger volume.



◁ Liquid water

In liquid water there are fewer hydrogen bonds. The molecules can get closer to each other, taking up less volume. The bonds break and reform, allowing the molecules to move around.



◁ Water vapor

In the gaseous form, the water molecules of water are independent of each other and can move around freely. Water vapor forms below the boiling point of water, while steam is technically vapor above 100°C (212°F).

Water is **densest at 4°C (39°F)**, which is the temperature at the bottom of the deepest ocean floors.

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